

Prateek Vashishtha

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EDUCATION

- Vellore Institute of Technology** Bhopal, India
Bachelor of Technology - Computer Science and Engineering; CGPA: 8.61 Sep 2023 - May 2027
Courses: Artificial Intelligence, Machine Learning, Human Computer Interaction, Data Structures, Algorithms, Databases
- Cambridge School Greater Noida** Greater Noida
Class XII (CBSE) July 2022
- Cambridge School Greater Noida** Greater Noida
Class X (CBSE) July 2020

SKILLS

- Programming & Core CS:** C++, Python, Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks
- AI & Machine Learning:** Machine Learning, Deep Learning Basics, LLM Integration, Prompt Engineering, NLP, Explainable AI
- Backend & Systems:** Python (Flask), REST APIs, System Design Basics, API Integration, Docker, AWS
- Data & Databases:** SQL, Pandas, NumPy, Data Pipelines, Data Modeling, Query Optimization
- Tools & Workflow:** Git, Postman, Power BI, Debugging, Rapid Prototyping, Experimentation

EXPERIENCE

- Indian Council of Agricultural Research - IISWC** Dehradun, India
Research Intern Dec 2025
 - Decision Support System:** Developed an AI-driven decision support system to assess soil erosion risk using geospatial, terrain, and rainfall data, enabling intelligent and scalable recommendations.
 - Backend for integrating Live APIs:** Built a Python-Flask backend integrating ML models with live terrain and rainfall APIs to simulate real-time decision workflows and automated recommendations.
 - LLM Integration for Explainer Agent:** Designed and integrated an LLM-powered explainer agent to generate human-readable insights, improving interpretability and user interaction with model outputs.

PROJECTS

- UPI Transaction Data Analysis Dashboard | Power BI, SQL, Python:**
Built a data processing pipeline to analyze large-scale transaction datasets and uncover behavioral patterns. Applied structured querying and optimization techniques to extract insights efficiently. Designed dashboards to simulate real-world financial analytics systems.
Tech: Python, SQL, Pandas, Power BI Sep 2025
[GitHub]
- Employee Attrition Prediction System | ML, Python, SQL:**
Developed a machine learning pipeline to predict employee attrition using structured HR data. Applied feature engineering, model evaluation, and explainability techniques (SHAP) to interpret predictions. Simulated a decision-support system for real-world deployment.
Tech: Python, Scikit-learn, SQL, Pandas, SHAP Nov 2025
[GitHub]
- AI-driven Soil Conservation System | ML, LLMs, Backend:**
Built an end-to-end intelligent system combining ML models with geospatial data to predict soil erosion risk. Integrated a Flask backend with real-time data inputs and developed an LLM-based module for generating interpretable recommendations. Simulated an AI agent workflow combining prediction, reasoning, and user interaction.
Tech: Python, Flask, Pandas, SQL, LLM APIs Dec 2025
[GitHub]

CERTIFICATIONS

- Oracle Certified Data Science Professional - Sep, 2025
- Deloitte Data Analytics Job Simulation - July, 2025
- Human Computer Interaction(Elite), NPTEL - June, 2025
- Introduction to Machine Learning, NPTEL - Sep, 2024

POSITIONS OF RESPONSIBILITY

- Lead, Public Relations and Outreach** Aug 2025 – Present
UI/UX Club
Spearheaded product-focused initiatives including hackathons and mentorship programs for 200+ students, emphasizing user experience, problem-solving, and real-world applications.:
- Core Member, Content Team** Nov 2024 – Mar 2025
AI Club
Contributed to Generative AI initiatives by managing content strategy, designing engagement campaigns, and promoting awareness of AI tools and applications.: